PT6: "Reliable Knowledge can lack certainty." Explore this claim with reference to two out areas of knowledge.

The symbiotic relationship between certainty and reliability lends itself to determining how valid and legitimate a knowledge claim is. A knowledge claim that is reliable is defined to be one that is repeatable, reproducible and tends to truth – this being evaluated on the basis of corroboration, consensus, pragmatic applicability and falsification. Given reliability's need for there to be external validation, it begs the factor of certainty to determine a claim's inherent value and legitimacy. In the case of this essay, a claim's certainty will be defined as the incorrigible, the indubitable and the unquestionable - specifically alluding to Reed's definitions of certainty as psychological certainty and epistemic certainty. "Psychologically certainty defined as supremely convinced of its truth" and largely influenced by strong personal beliefs and emotional confidence (Reed). "Epistemic certainty defined as a belief with the highest possible epistemic status" which is based on validation by reliable standards (i.e evidence or proof) or justified by the methods of the AOK (Reed). Given the imbued understanding that reliability and certainty share a collaborative relationship to establish if a knowledge claim is legitimate or not, it begs the following question: How much does reliability require certainty in the process of knowledge validation? Therefore, this essay will look at 3 perspectives – reliability will tend to certainty to a large extent, knowledge can be uncertain but reliable, and knowledge can be certain but unreliable depending on the context. These will be explored in the AOKs of Natural Science and History.

In the area of **Natural Sciences**, a **reliable knowledge claim tends to certainty to a large extent**. Given the empirical nature of natural sciences, epistemic certainty is

considered to establish the reliability and the legitimacy of a scientific knowledge claim- through quantifiable methods like experimentation and scientific consensus. For instance, Newton's laws of motion explain the abstract concept of gravity based on a series of repeatable experiments that yielded the same results across various objects - from an apple to large, bulky objects. Alongside the corroboration amongst the scientific community, scientists validated the logic behind theory of Newton's Law of gravity through applications such as space travel, escape velocity, period of planets as well as achievements like the 1969 Apollo rocket launch ("July 20, 1969: One Giant Leap For Mankind"). Newtonian physics proves to be reliable when developing laws of motion and gravitation based on repeatable, experimental consensus and its pragmatic applicability in a non-quantum realm. Given its strong epistemic experimental proof, corroboration and quantifiable justifications, it tends towards a large degree of certainty. Therefore, scientific knowledge claims constitute reliability to tend to certainty to a large extent.

However, it is notable that natural science knowledge claims can be **reliable but uncertain**. This presents itself when a knowledge claim can be pragmatically applied, but the construction of the claim itself is not repeatable or reproducible. For instance, the Heisenberg's Uncertainty Principle. The Principle itself defies the rule of the co-dependent relationships between particle location and its momentum – by stating that the certainty of knowing a particle's location is inversely related to the certainty of its momentum (Homer, and Bowen-Jones). Given this inverse relationship, the Principle defies other quantum physics experiments that provide probable explanations between a particle's momentum and/or location (Hilgevoord, and Uffink). This contradicts shared knowledge about these two factors, making the

Principle counterintuitive and defying the rules of quantum physics – making it theoretically an uncertain piece of knowledge. However, it supports pragmatic applications and achievements in the scientific field such as low noise technology (Gureyev et al.). Since the Principle is inherently uncertain as it defies the rules of epistemic certainty (i.e it does not follow the requirements of empirical justification and logical reasoning), but satisfies pragmatic applications, it exemplifies the nature of some claims that can be reliable but uncertain in the Natural Sciences.

On the contrary, scientific knowledge claims can be **unreliable but certain**. This is considerably plausible in the area of pseudo-science. The nature of certainty in this context tends to psychological certainty, which depends largely on emotional justifications. This can be seen in vaccinations. Vaccines, despite being proved to counteract threatening diseases, have been pointed out to induce other health related problems in individuals. One such problem that has been identified, is the development of autism by Wakefield's theory (Boseley). Theoretically the argument presented was flawed under the premises that the claims made could not be repeated in experiment and reproduced consistently (Boseley). However, it did consider the stories of patients who had been vaccinated, thereby making the emotional correlation to the theory very strong (Boseley). This supports the premises of psychological certainty but fails to meet the standards of reliability for any knowledge claim and a general scientific consensus in support of the claim. Therefore, in a situation where context and applicability of a knowledge claim is cherry picked and used, the claim possesses a high degree of psychological certainty but remains unreliable.

Apart from **Natural Sciences**, these three perspectives can also be seen in the AOK of **History**. The knowledge claims in history also follow a similar pattern to that of the natural sciences whereby the reliability of a claim tends largely to certainty. In the case of historical methodology, a large importance is given to historical data and source consensus.

History predominantly adopts pluralistic triangulation methods of verification and corroboration, whereby primary and secondary sources of data are used and cross-checked in establishing the legitimacy of historical claim. For instance, legitimacy of the historical claim of the fall of the Berlin Wall – a significant moment in the history of the Cold War. The event was corroborated by primary data surrounding the economic aftermath in Germany and Eastern Europe, eyewitness accounts that narrate details of the fall, political records that documented the key events leading up to the fall (Fisher). These sets of data, many theoretical frameworks were adopted by historians to corroborate and validate the certainty of the fall, therefore subscribing to pluralistic methods of historical verification, and in turn, epistemic certainty. Therefore, the corroboration and paralleling historical theories about the Berlin wall support and validate its reliability as a historical event , but also enables the claim to tend towards certainty on the basis of logical sequencing of events preceding the fall and its quantitative impacts on society to classify it as an important and certain historical event (Fisher). However, given the subjective nature of historical methodology, it is also probable to consider that a historical claim can be reliable but uncertain. This is more prominent in the case of new and emerging pieces of evidence. The fall of the Berlin Wall was monumental in liberating East Germans from a communist state to seek democracy in the West. Even though the majority of

historical accounts and data speak for the benefits that came with the fall of the wall, post-war evaluations of Germany and its citizens did show new evidentiary discrepancies in the quality of living in a communist regime as well as a democracy (Stephenson). For instance, post-war, it was quantified that there were some Germans who believed that their lives were better in a communist state, due to the inability to match income levels with West Germans and other economic factors (i.e social mobility, economic productivity etc) (Stephenson). This created a subtle yet noticeable sentiment that there were Germans who believed that the fall of the Berlin wall was not beneficial in liberating Germans to get a better life (Stephenson). These sentiments were expressed in mostly personal beliefs and stories, coupled with some economic statistics (Stephenson). Hence, the legitimacy of the claim amongst Germans that the wall ensured liberation for Eastern Europe loses some of its certainty as the evidence and nationalist ideologies of the aftermath of the fall, conflict each other. However, the reliability that the claim was a turning point in the Cold War remains.

On the other hand, it is possible for a historical claim to **warrant certainty but not reliability**, especially when in the context of revisionist perspectives. In the case of the Berlin Wall, this becomes evident when considering the claim in a 21st century context. As modern historians evaluate the various sources (both primary and secondary), the conflicting nature of the claims made about the fall of the Berlin wall and its aftermath reduce the reliability of the claim that the fall of the Berlin wall was beneficial in the liberation of East Europeans (Fisher). Given the combination of psychological certainty based on survivor stories as well as epistemic certainty based on quantitative data outlining the effects of the fall, the claim remains highly

certain but poses as unreliable. The subjective nature of how the different perspectives of the claim are constructed, the validity of the claim in the larger argument of the Cold War becomes unreliable. Hence, the contextual understanding and evaluation of historical claims can result in the claim having high certainty but little reliability.

In conclusion, the relationship between reliability and certainty of a knowledge claim does depend on contextual analysis of the claim. That being said, the requirement for certainty for reliability varies to different extents depending on the type of certainty being associated with the claim in question. Only when this is established, can one use the factors of reliability and certainty to deduce the validity of a knowledge claim in the context in which it is being evaluated. 100% certainty is impossible. The relationship between reliability certainty can also be used to understand the extent of validity a claim can have, and by extension, its accuracy and relevance in shaping knowledge in a certain paradigm. Therefore, the question of the co-existence or mutual exclusivity of reliability and certainty will determine what and how knowledge is perceived and therefore how knowledge systems and paradigms change.

(1600 words)

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